🌐 Introduction to VLANs (Virtual Local Area Networks) (10.30.2026)

> **Course:** CCNA 200-301 | **Source:** Jeremy's IT Lab
> **Tags:** #CCNA #Networking #VLANs #Switching

📡 1. Core Concepts: LANs & Broadcast Domains

Before understanding VLANs, we must understand the foundation of local networks.

- **LAN (Local Area Network):** A network that interconnects devices within a limited area (like a home, office, or building).
- **Broadcast Domain:** A logical division of a computer network in which all nodes can reach each other by broadcast at the data link layer.
  - *Rule of thumb:* Devices in the same broadcast domain receive each other's broadcast frames (e.g., ARP requests).
  - Routers **break up** broadcast domains. Switches **do not** (by default).

🧩 2. What is a VLAN & Why Use It?

A **VLAN** is a logical grouping of devices on one or more LANs that are configured to communicate as if they were attached to the same wire, when in fact they are located on a number of different LAN segments.

Why segment the network with VLANs?

- 🔒 **Security:** Isolate sensitive traffic (e.g., HR vs. Engineering).
- ⚡ **Performance:** Reduces broadcast traffic (smaller broadcast domains = less noise).
- 💰 **Cost Efficiency:** Reduces the need for extra hardware (you don't need a physical router for every subnet).
- 🛠️ **Flexibility:** Easily move users between groups without changing physical cabling.

⚖️ 3. Layer 3 vs. Layer 2 Segmentation

- **Layer 3 (Subnets):** Traditionally, routers are used to separate networks into different subnets. This separates broadcast domains.
- **Layer 2 (VLANs):** Switches can perform this segmentation internally using VLANs. This allows you to create multiple broadcast domains on a single physical switch.

🛠️ 4. Configuration & Defaults

- By default, all switch ports belong to **VLAN 1** (the Default VLAN).
- **Commands:** `vlan [id]`, `name [name]`, `interface [type]`, `switchport mode access`, `switchport access vlan [id]`
- **Verification:** `show vlan brief`

🧠 Quiz Time! (Test Your Knowledge)

📝 Text-Based Quizzes

Q1: What happens if you try to assign a switch interface to a VLAN that doesn't exist?

- a) The command will fail.
- b) The switch will create the VLAN.
- c) The interface will be disabled until you create the VLAN.
- d) All VLANs exist by default.

> **Answer: b) The switch will create the VLAN.**

Note: In modern Cisco IOS, if you assign a port to a non-existent VLAN, the switch automatically creates it in the VLAN database.

Q2: You create VLANs 10, 20, and 30 on a Cisco switch. How many VLANs will be displayed in the output of the `show vlan brief` command?

- a) 3
- b) 5
- c) 8

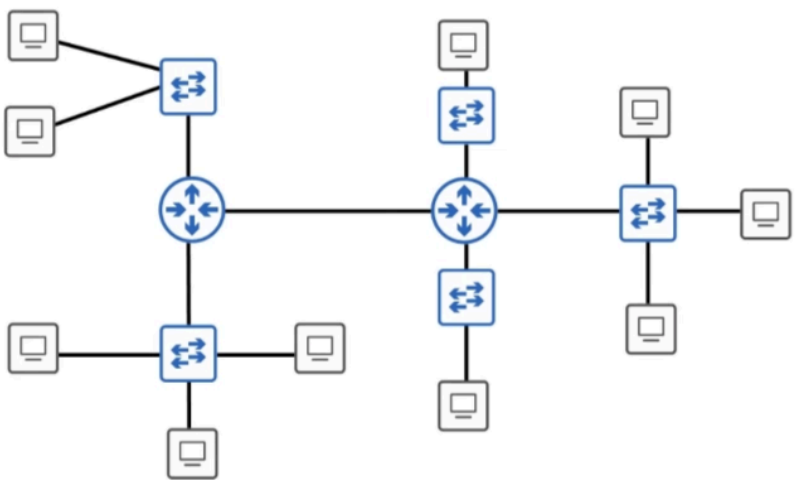
- d) 10

> **Answer: c) 8**

Explanation: You created 3. However, Cisco switches automatically have VLAN 1 (Default) and VLANs 1002, 1003, 1004, and 1005 (Token Ring/FDDI defaults). 3 + 1 + 4 = 8.

🖼️ Diagram Quizzes

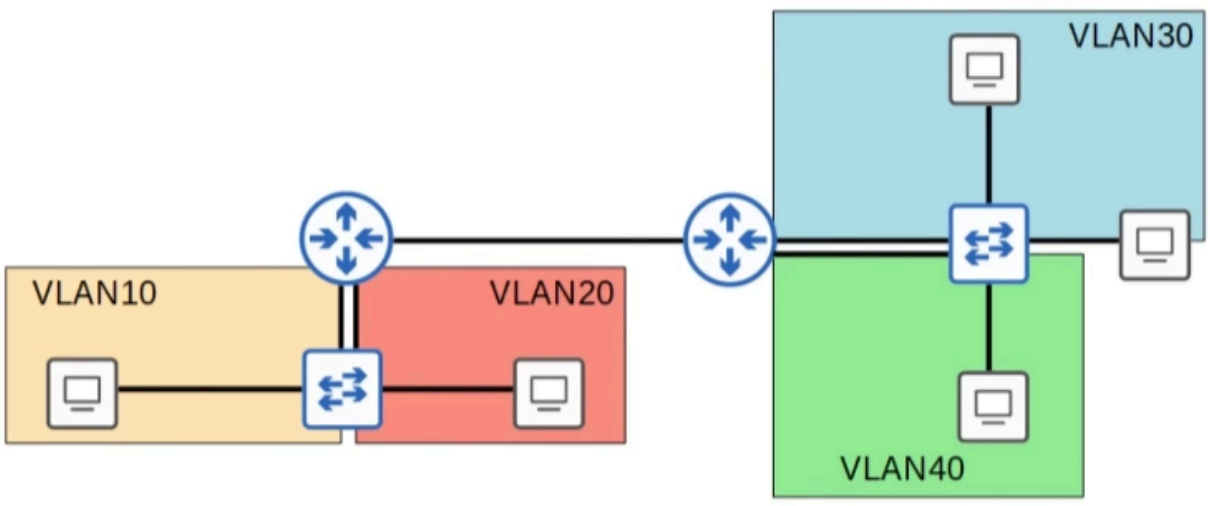
Quiz Image 1: Basic Network Diagram



Question: How many broadcast domains are shown in this network diagram?

> **Answer: 6 Broadcast Domains Explanation:** The circles with arrows pointing outward are **Routers**. The squares with arrows are **Switches**. A router interface acts as the boundary for a broadcast domain. Let's count the distinct segments separated by the routers:
>
> 1. Top-left switch + its 2 PCs
> 2. Bottom-left switch + its 3 PCs
> 3. The point-to-point link between the left router and middle router
> 4. Top-middle switch + its 1 PC
> 5. Bottom-middle switch + its 1 PC
> 6. The entire right side (rightmost switch + its 3 PCs connected to the middle router)
>   *(Since switches do not break up broadcast domains, all PCs plugged into a single switch without VLANs are in the same domain).*

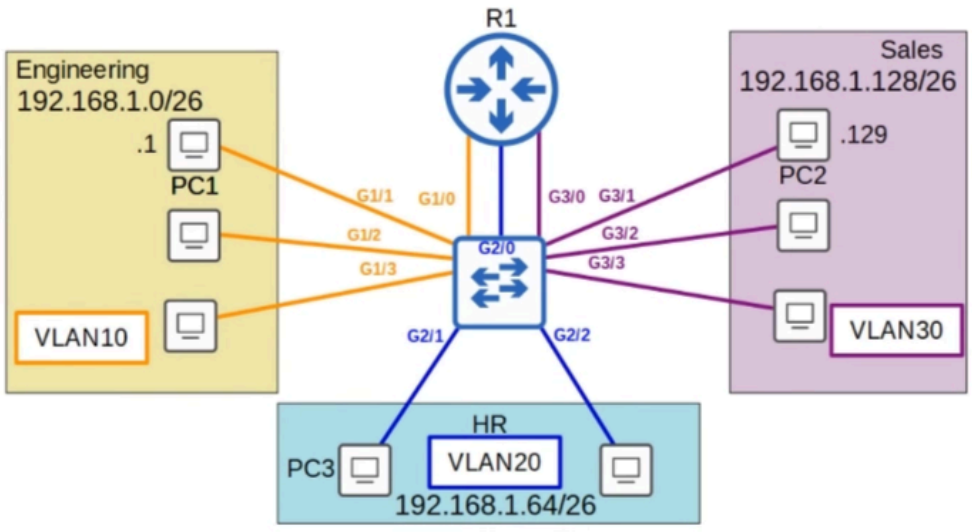
Quiz Image 2: VLAN Segmentation



Question: Looking at the color-coded sections, how many broadcast domains are shown?

> **Answer: 5 Broadcast Domains Explanation:** VLANs are used to logically separate a physical switch into multiple smaller broadcast domains. Additionally, router interfaces always bound a broadcast domain. Let's count them:
>
> 1. VLAN10 (Yellow)
> 2. VLAN20 (Red)
> 3. VLAN30 (Blue)
> 4. VLAN40 (Green)
> 5. The point-to-point link connecting the two routers.

Quiz Image 3: Broadcast Propagation



Question: If PC3 sends a broadcast message, how many devices will receive it?

> **Answer: 3 Devices Explanation:** A broadcast message stays within its specific VLAN/broadcast domain. PC3 is located in the **HR VLAN20** broadcast domain.
>
> 1. The **switch** since it receives the message from PC3.
> 2. The **other PC** in the HR VLAN20 box (The switch will flood the broadcast out to this device).
> 3. **Router R1** (The router is connected to the switch via the blue link, which acts as a trunk carrying VLAN20 traffic. The router's subinterface for VLAN20 will receive the broadcast).
>
> *Note: The switch itself also processes the frame to flood it, but in standard CCNA counting, we look at the sender, the end receivers, and the router interface.*